

DVD Player Upgrade Mod

Tuesday, August 11, 2026 2:05 PM

My rig came with the rear entertainment audio system. When I first got it, I thought it was the DVD player option since I didn't do that much research and from a distance, it looks like it. Come to find out, Toyota made 2 versions of the "rear entertainment" - one for just audio and the other for DVD. Mine was the audio one. It seems less common because most of the posts you see out there and other rigs for sale, it seems more like the rear DVD option is seen.

Here is a view of the rear audio entertainment system. Essentially, it's a small screen that supposedly turns on with a remote that came with the vehicle and you could tune into your own radio station from the backseat. There are 2 headphone plug-ins accessible from the backseat as well, near the air control, where you can plug in wired headsets to listen to the different radio station. I've yet to see an example of one that still works. Mine didn't come with the remote and I could never figure out how to get it to work. I believe it only works with the factory headset too. Wild.



I did some digging and it's the same frame as the DVD player screen. That got me thinking... Maybe I can switch it out for the DVD player? I didn't want to just do the OEM Toyota, 20+ yr old screen that has to work with the center console player. I wanted a polished, new-age screen that is today's tech for HDMI and clarity.

- There are 2 phillips head style screws behind the map light diffuser - just pop off the diffusers and you'll see them
- Take out those screws and then you can start from the switch side and pop down the outer cover then the back part will come down too - leaving the main frame with the center entertainment screen part in tact

Where could I get an inexpensive screen, that is barebones, where I can make it fit in the OEM case... This is a great candidate for AliEx! After much measuring the OEM. Here is the screen I went with (<https://www.aliexpress.us/item/3256808147329677.html>):



I do have younger kids that have enjoyed this on longer rides so far, but they also have tablets and would prefer those over this most of the time. I just like the idea that I was able to add a newer unit into a retro setup and make it look OEMish.

Let's get started...

Parts Needed:

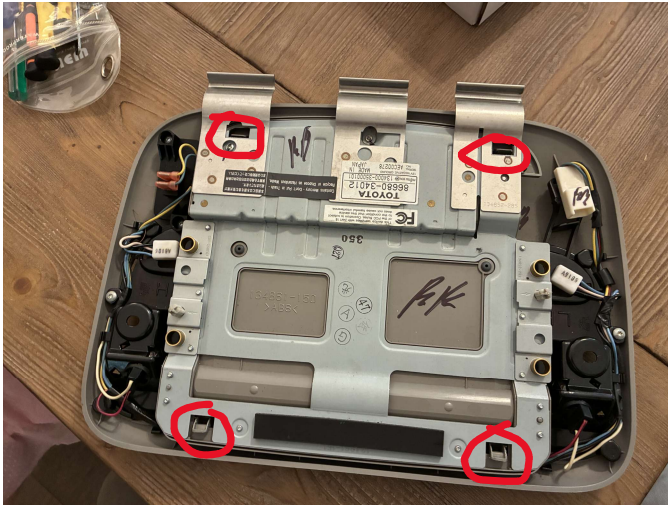
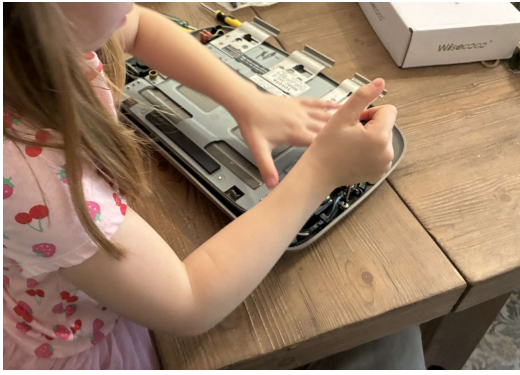
- Wisecoco 7" monitor (linked above)
- A way to connect wires - your choice (I chose to use Wago wire nuts: <https://a.co/d/0c4UswxE>)
- Wireless HDMI transmitters (<https://a.co/d/08sJUXy7>)
 - You could run an HDMI wire through the headliner down one of the pillars to wherever you wanted but I didn't want all that work
- Android TV box of your choice - anything with an input for USB
 - I connect the usb with downloaded media to the box to then play over the monitor
- 12v relay to be able to switch on the monitor when it's pulled down from ceiling (<https://a.co/d/0gwoLokZ>)

Steps:

You can get the full DVD assembly from eBay if yours didn't come with it or if you do have the DVD assembly, then you can skip this step. This is how mine came from eBay



Start the disassembly - I had my oldest help me out! She loves projects with dad

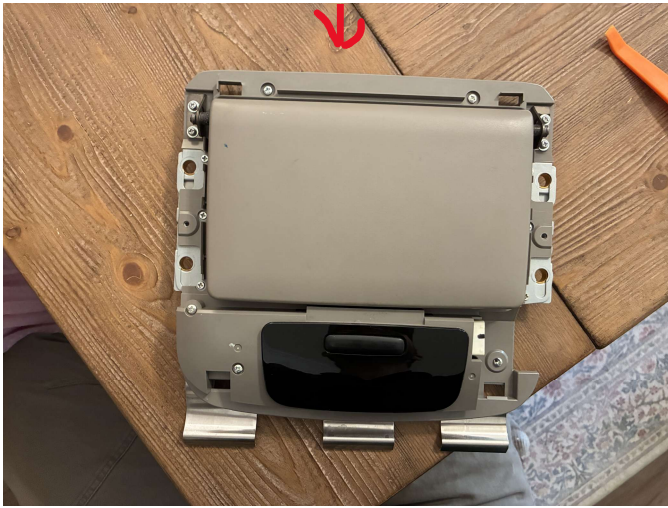


There are some pressure clips holding the main DVD entertainment screen assembly to the frame
You just need to pop down those clips and it comes apart in 2 separate pieces



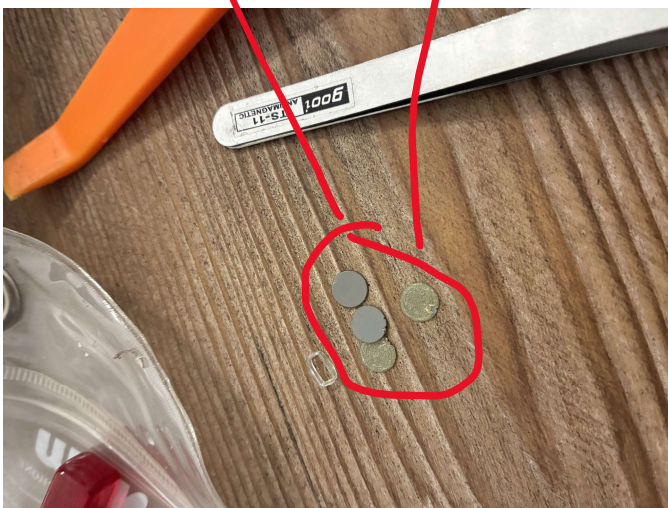
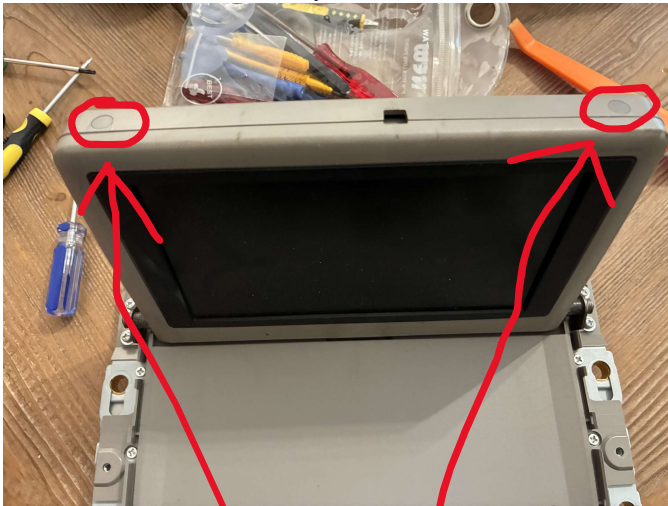
Fun fact, this surround frame is the same for rear seat AUDIO entertainment as well as the DVD entertainment





Now you need to get down to the bare components of the OEM player monitor screen, so we'll focus here

You'll want to take off all the thin covers for each of the screws - I used a little pick tool (seen in the pics). There should be 4 total, but I just have 2 circled below for example



Then you'll see the little screws behind



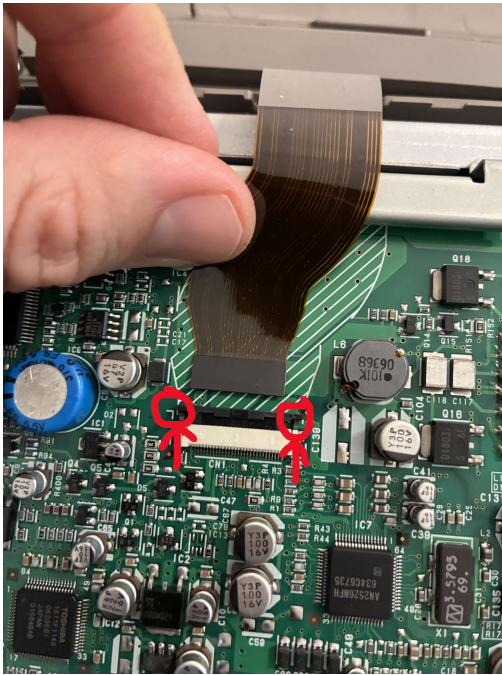
You can also go ahead and take out the 4 screws (2 each side) holding down the screen to the assembly, just be careful as there is a ribbon cable connecting the screen to the circuitry and you don't want to tear that



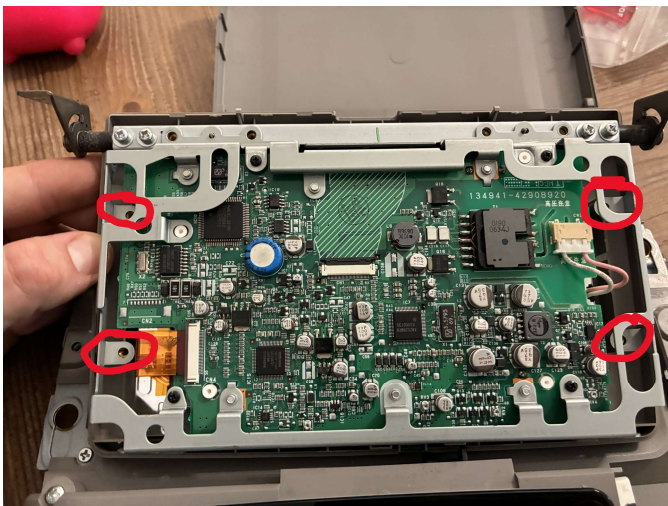
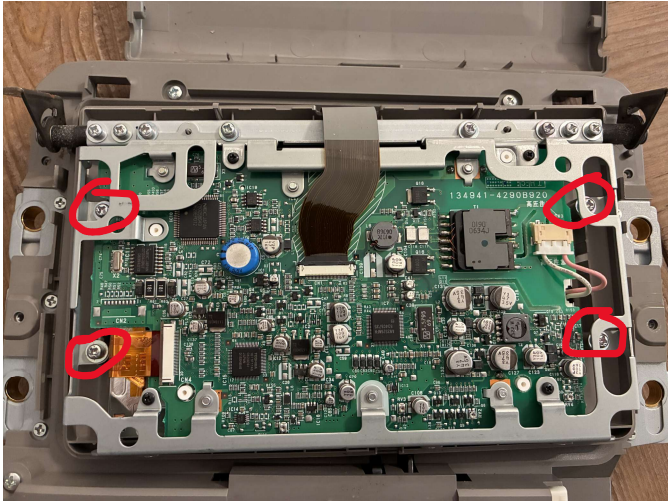
Then you should be able to pop off the back cover and see what's inside...



You can disconnect the ribbon cable to pulling slightly on the ears of the black connector on both sides then sliding the cable out



Now you can take out the 4 screws that hold this metal frame + screen to the front cover face



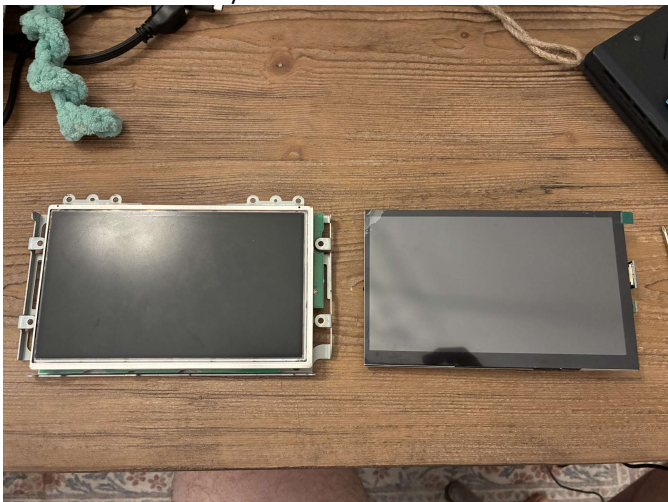
Then the whole screen comes out from the front frame just like this

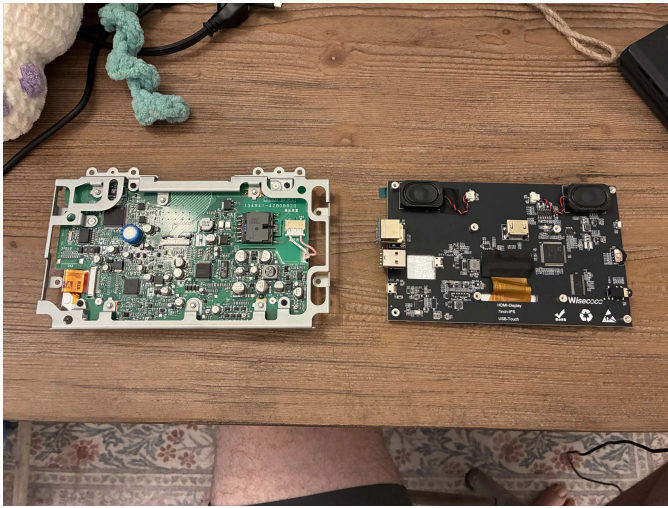


Here's a pic of the front frame



Here are some side-by-side shots of the old vs new





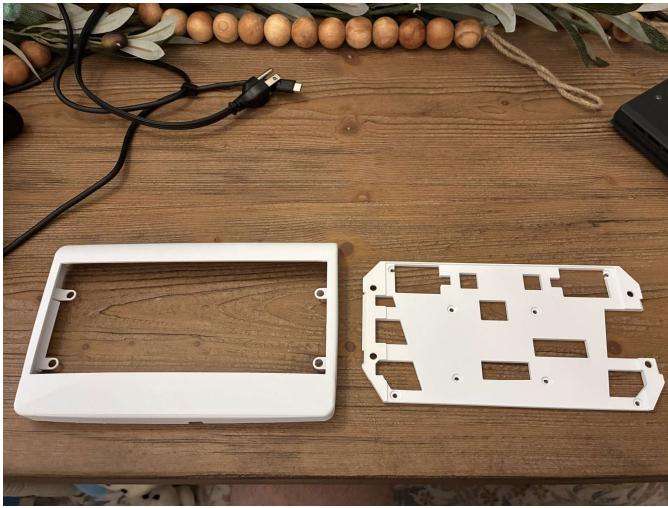
Now, you wonder, "how do I mount this new screen in the cavity of the OEM parts?"

I took on some painstaking measuring of the original mount locations from the OEM front face and the new monitor and figured out how to adapt the 2

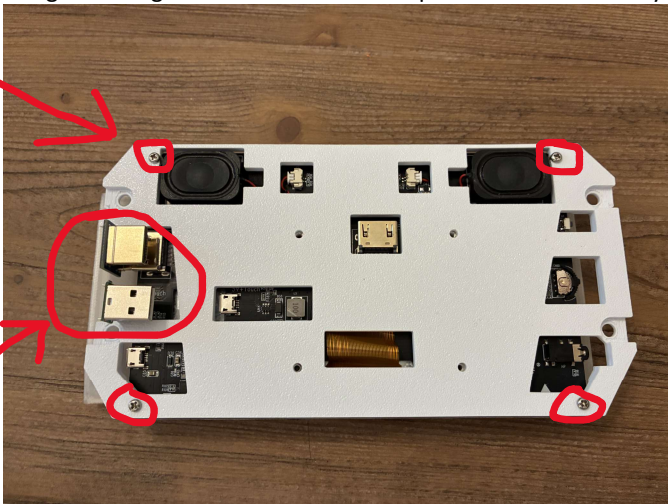
Then I had to figure out how to mount all the "guts" aka power cords, HDMI wireless adapter, etc. in the cavity behind the screen AND have it all concealed... I was originally going to use the OEM back cover as well but it was too tight for everything to fit in there.

So, here we are again, I measured all the OEM features of how the back cover mounted but modified it to be deeper and had a semi-open back to be able to mount a cover to it after I was done with assembly to keep everything in

This is what I came up with for the printed pieces. These are just prototypes but ended up being the design I went with

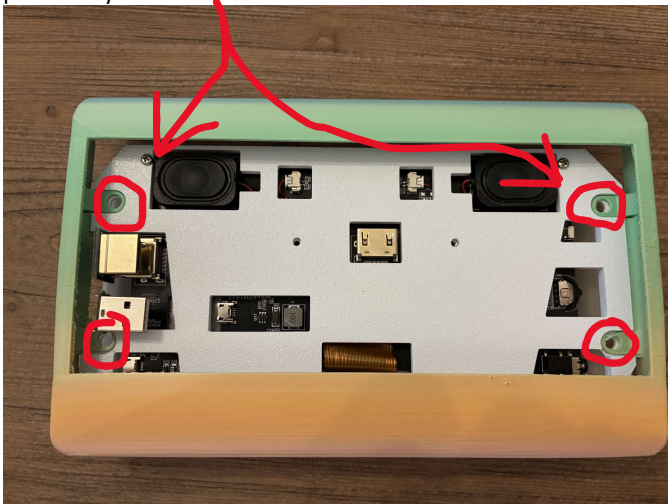


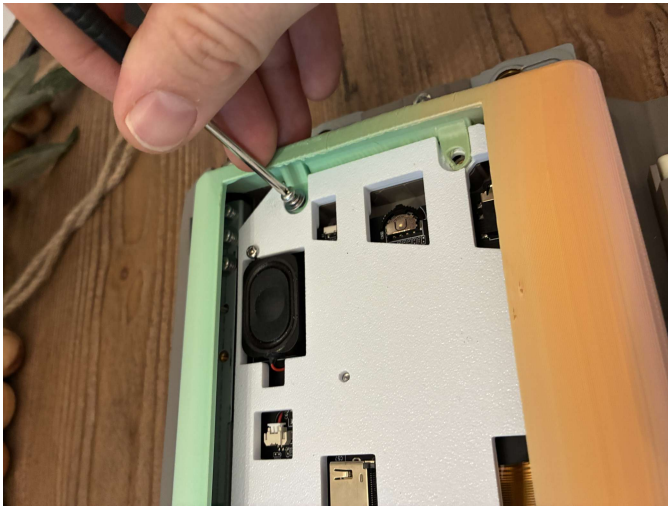
The new screen mounts to these spots here (I didn't use the other 4 holes you see in the pic but added them in the design thinking I would use them to help mount - it was sturdy enough with the 4 larger hole patterns)



These 180 deg ports come with the monitor kit from AliExpress and came in super useful to help route the connections back into the open space or else I would have had to buy tight right angle cords

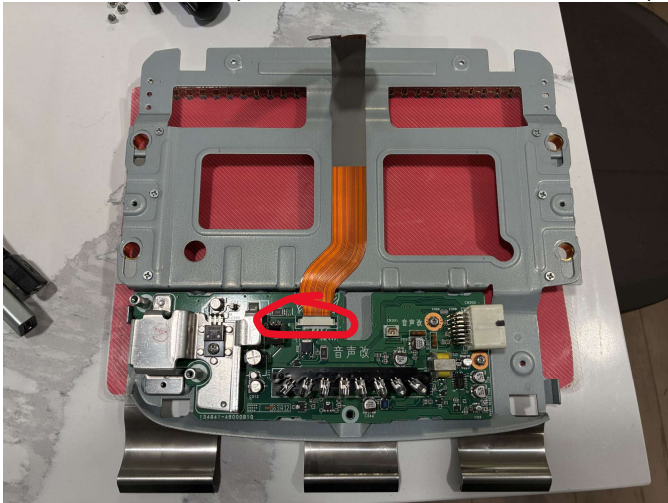
Then my new, modified back cover follows the same form as the OEM one but is deeper and has different connection points as you see here



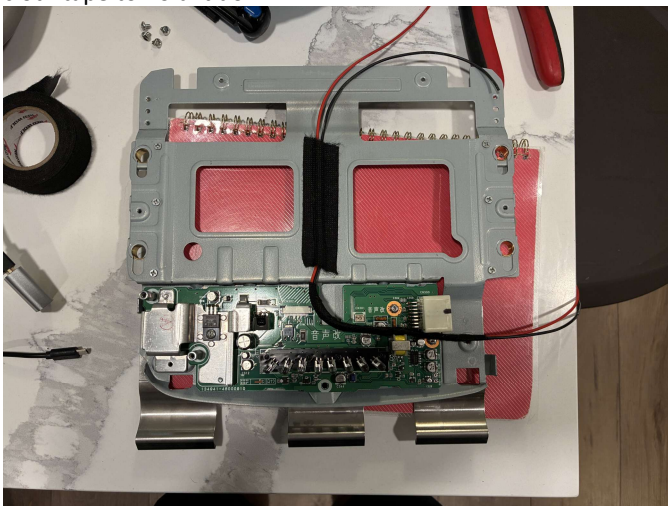


Now this subassembly is ready for the new screen, we need to doctor up the other base piece

You will need to totally remove the ribbon cable now - same process as before with the clip



Then you'll want to run you a length of red and black (pos and neg) wires through this cavity area - I just used some soft cloth tape to hold it down



Now, you can set your new screen assembly up to the base and connect those 4 screws for the swivel arm to the frame - you'll also want to fish the red and black wires up through the cavity you see here and tuck it in nice and neat





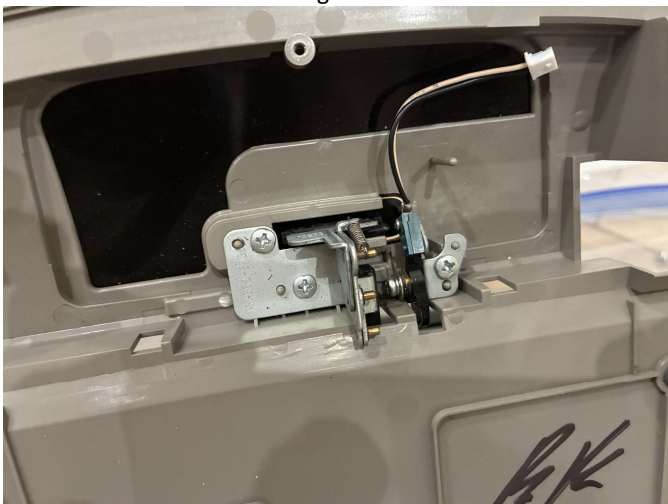
I could have just added a switch somewhere but that would have been more lengthy wiring and not as cool

I decided to dig into the OEM setup a bit more and realized it has a momentary switch that recognizes when the screen is up or down - I don't know what this would have tied to for the original logic of the car, but I decided to use it

Here is the switch I'm talking about - it's on a spring and activates the switch



Here is what the inner workings look like





Only issue is it's normally open when the screen is closed then the switch is off when the screen is open - backwards from how I need it...

No need to fear, a 12v relay to the rescue!

I had to do some research since I've never really used a relay. I found out that if you wire it up a certain way that it will reverse the action of a momentary switch. Perfection

I, unfortunately, did not get pics of this next part as I was wanting to just get this done. However, I had some help with AI and this is how I wired it up

✓ COMPLETE WIRING (Relay + Switch + Screen)

Relay Pins

- 85 → Ground
- 86 → Switch output
- 30 → 12V power source
- 87a → Screen + (power input)
- 87 → leave unused

Switch Wiring (this is the part you were missing)

Your switch has 2 wires. You must turn it into a real 12V trigger circuit:

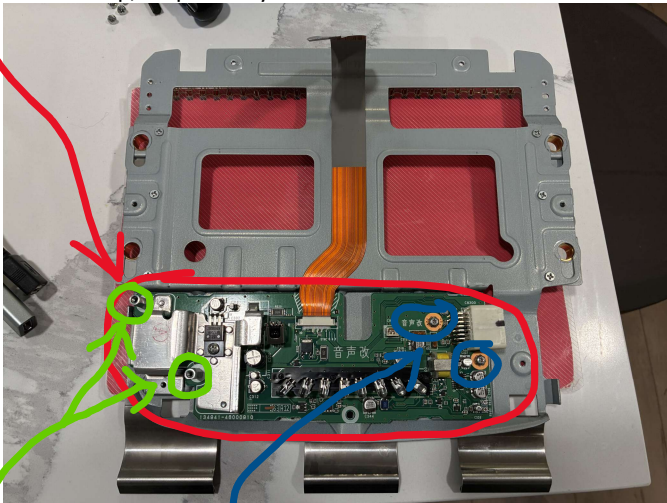
12V → SWITCH → pin 86

So:

- One switch wire → 12V (tap into ACC or constant 12V)
- Other switch wire → pin 86 on relay

Remember the electronics that were all in the frame part? This is the IR reader for the OEM remote and the circuitry for sending everything through the ribbon cable to the OEM screen.

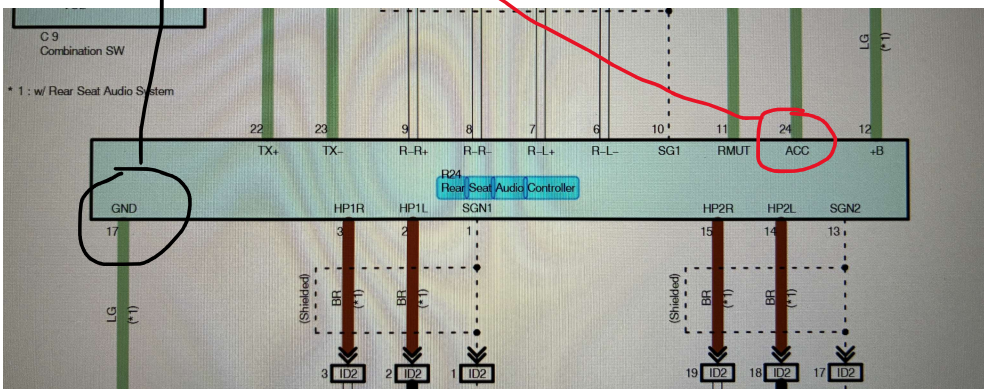
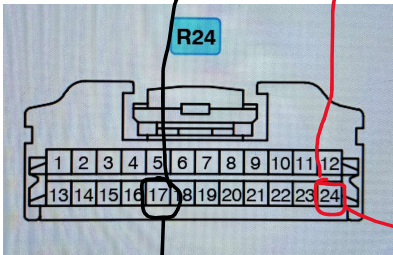
I used that whole area to house the relay and wiring to the small OEM switch for power on and off when screen down and then up, respectively.



There are various screws on top and bottom (didn't get a good pic from the bottom end but you'll find them) AND you have to thread out this female screw housing from the board to free it up in order for it to come apart

Now, head back to the car and we'll dig into how I stole power and ground from the OEM connector

If you look at the back end of the connector from OEM, like my pic below, the bottom left (pin 24) is power +12v when ACC or ON for switch and on the bottom row towards the middle (pin 17) is - ground



Service Hints

S9 (A), S10 (B) Stereo Component Amplifier
 (A) 1, (A)10-Ground : Always approx. 12 volts
 (B)12-Ground : Approx. 12 volts with the ignition SW at ON or ACC position and acc cut relay not operated
 (A) 15, (A) 16-Ground : Always continuity

R30 (E) Radio and Player with Display
 (E) 1-Ground : Always approx. 12 volts
 (E) 11-Ground : Approx. 12 volts with the ignition SW at ON or ACC position and acc cut relay not operated
 (E)20-Ground : Always continuity

R24 Rear Seat Audio Controller
 12-Ground : Always approx. 12 volts
 17-Ground : Always continuity
 24-Ground : Approx. 12 volts with the ignition SW at ON or ACC position and acc cut relay not operated

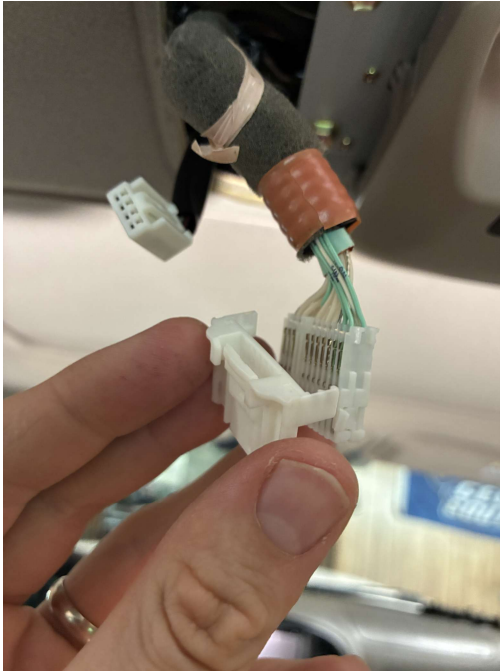
Yes, I could have used pin 12 for constant +12v, but I didn't want to risk my relay acting up and then the screen to be constantly on when I don't want it to and run my battery down flat

We only plan to use the screen and player when on trips and the car would be on anyways, so that's the route I took. You could certainly swap out pin 12 for constant +12v if you wanted

I'm not a fan of splicing into wires if I can help it, so I decided to disassemble the connector to free up the metal pins inside that I needed.

The top part clips on from the sides and you have to get a tool to free up the sides to get it off.

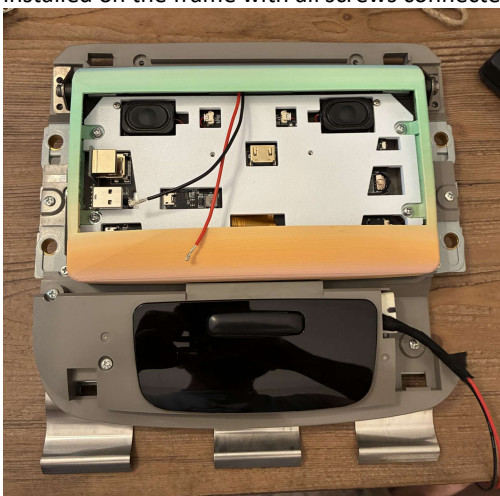
Here is a pic of me disassembling it then you can separate the layers (top set to bottom set) and remove out the wire with metal connector end that you need and then put the rest back together



Now we'll keep those loose wires until we're ready to make connections

Now we're ready to do the final steps on connecting all the wiring at the screen itself and then install in the car!

Back to the screen assembly, you should have something that looks somewhat like this below - it'll be the screen back installed on the frame with all screws connected to hold everything down and + and - wires that are routed through

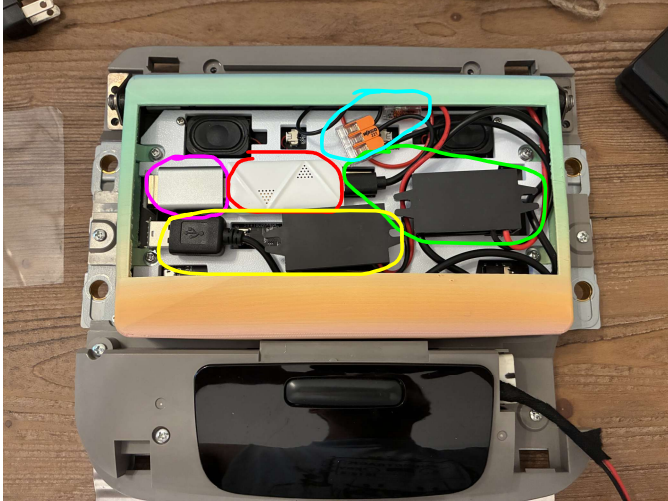


Here's everything installed with all connections made you should see:

- USB-A Step Down converter 12v to 5v

- For main power to the screen
- **USB-C Step Down converter 12v to 5v**
 - For main power to the HDMI wireless receiver - found out I may not have needed this since the HDMI converter may get enough power through HDMI connection - still messing around with this
- **2x Wago connectors** splicing all + and - power connections
- **HDMI female to female connector**
 - Needed to buy this since I didn't have one laying around to connect the male HDMI receiver to male 180deg connector from the screen
- **HDMI receiver** from the Amazon kit

It's tight! But it'll all snugly fit in there especially with my flexible cover for this back panel



Now, I just installed my cover on top of that

This is clear TPU that I printed this out of - I wanted some flex to help install and thought some translucence would be fun to see the electronics behind

Remember, this whole back cover combo with cover is not what I play on going with full time it's just for now while I was testing everything - I'll probably go back and print a close color match to the tan OEM color or just pick black for the final version



Now, here it is ready to install in the car!

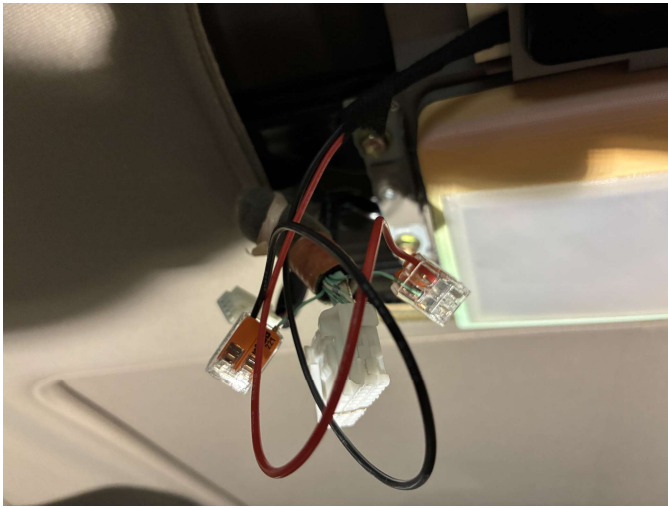
I got multiple views - you can see it doesn't really bump out much more than the OEM one did - it also doesn't matter, since this is protruding into just open space of the car



So, at this point, I just installed the screen assembly like you see it above in place of my original OEM rear audio entertainment system (or if yours came with the OEM DVD player then it would be putting it right back in place as it was, but upgraded)

All that's left after you follow the steps of installing the hardware for the screen assembly to the roof area of the car, is making the last wiring connections for power from the de-pinning we did earlier

I used the Wago connectors again here - just a simple + power and - ground is all that is needed at this point



Here it is after power connections made and before final outside trim + map lights have been installed



Here it is in it's final (until my new printed color change of case and cover are done) state!





Now, the whole reason for getting the specific Android TV box from earlier is for multiple reasons - I don't have a way to have internet connection on the road (don't pay for the hotspot feature) so we can't use the online version of Android

The great thing about these Android TV boxes that come with USB connections is that you can plug in any offline videos or media you want from USB and the box will read and play it - they don't rely on internet connection for all features

Also, having the Android operating system, I can link wireless bluetooth headphones to the box and have the media play through that - the AliExpress screen comes with cheesy built in speakers that work, but as you can imagine, are garbage - I did not opt in to having the screen connected to my main head unit for the car to play through the car speakers because I couldn't think of a practical reason why I would have wanted that - we have this set up for the kids to watch their thing while we are talking in the front or listening to other music/podcasts

We've now come to the end of a VERY LONG writeup. I hope this was a fun read and may help others be creative in bringing our 1st gens up to this era!